

HELIOS Project: Comprehensive Contracts and Commercial Management Manual (FIDIC Yellow Book 1999)

1. Contracts Management Overview

The administration of utility-scale Solar Photovoltaic (PV) Engineering, Procurement, and Construction (EPC) projects demands an exacting standard of commercial governance. In the context of the 100 MW HELIOS Solar PV and 33/220kV Grid Substation (GSS) project in Uzbekistan, the Contracts Manager (acting as the COUNSEL agent) serves as the primary custodian of the project's commercial and legal baseline. Day-to-day responsibilities require the COUNSEL to synthesize schedule data, cost reports, and site records to formulate formal contractual correspondence, proactively protect the project against time bars, manage the variation lifecycle, and ensure cash flow remains unhindered. This role requires constant vigilance in translating physical site events into contractual mechanisms, orchestrating the outputs of claims engineers, quantity surveyors, and commercial coordinators to maintain the project's financial viability.

The FIDIC Conditions of Contract for Plant and Design-Build, First Edition 1999 (the "Yellow Book") is the industry standard for solar EPC projects where the contractor assumes comprehensive design responsibility. The Yellow Book differs fundamentally from the FIDIC Red Book 1999, which relies on employer-provided design and the remeasurement of estimated quantities. Instead, the Yellow Book is administered as a lump-sum contract where the contractor guarantees fitness for purpose based on the Employer's Requirements¹. It also contrasts sharply with the FIDIC Silver Book 1999 (EPC/Turnkey), which places almost all project risks—including unforeseeable ground conditions—on the contractor and removes the independent Engineer role entirely. In the Yellow Book, the Engineer acts as the Employer's agent but is strictly required under Sub-Clause 3.5 to make fair determinations based on the contract³.

A comprehensive mastery of the FIDIC Yellow Book 1999 requires the COUNSEL to possess intricate knowledge of its core clauses. Clause 1 establishes the contractual terminology and strict protocols for communications; Sub-Clause 1.3 mandates that notices be explicitly identified and delivered to the correct address⁴. Clause 4 delineates the contractor's general design and execution obligations, including the critical Sub-Clause 4.2 governing the provision and continuous validity of the performance security⁴. Clause 8 governs commencement, delays, and suspension, with Sub-Clause 8.4 providing the exclusive grounds upon which an Extension of Time (EOT) may be claimed⁶. Clause 10 manages the Employer's taking over of the works once they are substantially complete and have passed the rigorous Tests on Completion². Clause 13 manages the initiation and valuation of variations, while Clause 14 dictates the critical financial workflows, including applications for interim payments, advance payment amortization, and the application of financing charges for delayed payments².

Exceptional events, traditionally known as Force Majeure, are handled under Clause 19, and the rigorous claims and dispute resolution mechanisms are housed in Clause 20⁹.

The contract administration workflow relies on strict milestones, formal notices, and rigid time bars. The most commercially dangerous mechanism in contract administration is the time bar, located in Sub-Clause 20.1. This sub-clause explicitly states that a contractor must give notice of a claim for time or money within 28 days of becoming aware, or when it should have become aware, of the event or circumstance giving rise to the claim¹⁰. Failure to issue this initial notice results in the absolute forfeiture of the claim, permanently discharging the Employer from all associated liability. The Privy Council's 2024 decision in *Uniform Building Contractors Ltd v Water and Sewerage Authority of Trinidad and Tobago* strongly reaffirmed that Sub-Clause 20.1 operates as a strict condition precedent; neither the Engineer's informal knowledge of the event nor the Employer's subsequent termination of the contract circumvents this requirement⁹. Time bars are hazardous because they convert a mere administrative oversight into a total loss of substantive financial and temporal entitlement. Managing this risk requires the deployment of standardized delay event logs, mandatory daily site diaries, and an automated commercial calendar to trigger alerts the moment a disruptive event is identified on site.

2. Variation Order Management

A Variation Order (VO) represents a formal alteration to the agreed baseline scope of works, resulting in necessary adjustments to the Contract Price, the Time for Completion, or both. Under the FIDIC Yellow Book 1999, variations are initiated either by an explicit instruction from the Engineer under Sub-Clause 13.1 or via a request for the Contractor to submit a proposal under Sub-Clause 13.3¹². Instructed variations are overt and documented changes, such as the Employer formally requesting an upgrade in the capacity of the 33/220kV GSS autotransformers. Conversely, a constructive variation (constructive change) occurs when the Engineer issues a directive, approval, or rejection that is not labeled as a variation but fundamentally alters the scope, sequence, or method of performance. In the context of the HELIOS solar EPC in Uzbekistan, a constructive variation might arise if the Engineer rejects an approved module mounting structure (MMS) foundation design due to newly imposed seismic parameters not present in the original Employer's Requirements, thereby compelling the contractor to utilize deeper, more expensive piling.

The Variation Order process follows a rigid lifecycle orchestrated collaboratively by the COUNSEL and the CLAIMS agent. It begins with Identification, where the site or engineering team flags a discrepancy between the baseline scope and a recent instruction or site condition. This is immediately followed by Notification; the COUNSEL must issue a Sub-Clause 13.1 notice if it cannot obtain the necessary goods, or a Sub-Clause 20.1 notice if the change impacts time or cost¹¹. Valuation follows, governed by Sub-Clause 13.3. The CLAIMS agent prepares a fully detailed proposal encompassing direct costs, indirect costs, reasonable profit, and schedule impact⁹. The Engineer reviews the proposal and either approves it, rejects it, or proceeds under Sub-Clause 3.5 to determine the value. Upon approval or determination, the Engineer issues a formal VO, which the INVOICE agent subsequently incorporates into the next

Payment Application under Sub-Clause 14.3.

Valuation methods under a lump-sum FIDIC Yellow Book contract lack the detailed, unit-rate remeasurement framework of the Red Book⁹. Therefore, valuations are typically derived from four primary methodologies. First, utilizing Bill of Materials (BoM) rates, if the varied work is of similar character and executed under similar conditions. Second, using derived rates, where existing rates are mathematically adjusted to account for differences in site conditions or executed quantities. Third, dayworks under Sub-Clause 13.6, which compensates the contractor based on actual labor hours, equipment hours, and material costs expended, subject to daily verification and signature by the Engineer⁸. Finally, negotiated lump-sum pricing based on first-principles estimating, utilizing subcontractor quotes and material invoices. Strict adherence to notice requirements is paramount to preserve variation entitlements. If the Engineer issues an oral instruction on site, it must be confirmed in writing by the Contractor within two working days under Sub-Clause 3.3. If the Engineer fails to reject this confirmation within a further two days, it is deemed a valid Engineer's instruction. Furthermore, if a variation impacts the completion date, the 28-day notice under Sub-Clause 20.1 must be triggered independently of the valuation proposal under Clause 13, because the FIDIC Yellow Book requires Sub-Clause 20.1 compliance specifically for time consequences⁹. Contemporaneous documentation is the lifeblood of variation valuation. Records must include signed daily labor allocation sheets (daywork chits), equipment utilization logs, material delivery docketts (such as BNEF Tier 1 solar module import documents necessary for Uzbekistan customs clearance¹⁴), and dated site photographs comparing the original state to the varied state.

3. Extension of Time (EOT) Claims

The framework for claiming an Extension of Time under the FIDIC Yellow Book 1999 is established in Sub-Clause 8.4, which the CLAIMS agent must navigate with precision. Entitlement to an EOT arises strictly if the contractor is delayed by any of five specified grounds: (a) a Variation; (b) a cause of delay explicitly giving an entitlement to EOT under other sub-clauses (e.g., unforeseeable physical conditions under Sub-Clause 4.12); (c) exceptionally adverse climatic conditions; (d) unforeseeable shortages in the availability of personnel or goods caused by epidemic or governmental actions; and (e) any delay, impediment, or prevention caused by or attributable to the Employer or its personnel⁶. In the HELIOS project, ground (d) could be invoked if customs regulations abruptly change, such as the enforcement of BNEF Tier 1 exemptions delaying module clearance¹⁵, while ground (e) applies if the National Electric Grid of Uzbekistan (NEGU) delays granting access to the 220kV overhead line (OHL) route¹⁶.

The EOT claim process is sequential and strictly time-bound. Upon the occurrence of a delaying event, the COUNSEL must issue the Sub-Clause 20.1 initial notice within 28 days of becoming aware of the event. Subsequently, within 42 days of becoming aware of the event, the CLAIMS agent must submit a fully detailed claim encompassing the supporting particulars and the schedule analysis. If the delay has a continuing effect, this initial submission is legally considered an interim claim, requiring subsequent monthly updates until the delay ceases. A final claim must be submitted within 28 days after the end of the delay's effects¹¹. The Engineer

is then required to respond with approval, disapproval, or a request for further particulars within 42 days, ultimately proceeding to a Sub-Clause 3.5 determination.

Delay analysis requires deploying advanced forensic planning methodologies, the selection of which is heavily guided by the Society of Construction Law (SCL) Delay and Disruption Protocol (2nd Edition, 2017)¹⁹. The recognized methodologies include:

- **Time Impact Analysis (TIA):** Strongly recommended by the SCL Protocol for prospective or contemporaneous analysis, TIA evaluates the effect of a specific delay event by inserting it into a live, progressed schedule¹⁹. The CLAIMS agent defines a "fragnet" (a fragmented network of new activities mathematically representing the delay event). The analyst selects the accepted baseline or the schedule update whose data date is closest to, but preceding, the delay event. The schedule is progressed with actual as-built data up to the delay's start date. The fragnet is then logically linked into the schedule by assigning predecessors and successors to the delayed activity. The schedule is recalculated, and the difference between the projected project completion date before and after the fragnet insertion represents the discrete EOT entitlement generated solely by that event²¹.
- **Windows Analysis (Time Slice Analysis):** This method splits the project duration into consecutive periods (windows), typically mirroring the monthly schedule update cycle. The analyst tests which events drove the critical path inside each discrete window²². It is highly effective for complex utility-scale solar projects because it continually grounds the analysis in the contemporaneous reality of the project's progress, isolating overlapping delays month-by-month and preventing cumulative logic errors.
- **As-Planned vs As-Built:** A macro-level, retrospective approach that compares the original baseline schedule directly against the final as-built schedule, attempting to attribute the total variance to identified delay events²². This method is generally frowned upon by arbitral tribunals for complex EPC projects because it ignores dynamic shifts in the critical path over time and struggles to accurately isolate concurrent delays.
- **Collapsed As-Built (But-For Analysis):** A retrospective method that starts with the final, progressed as-built schedule and sequentially subtracts the duration of specific delay events (such as Employer delays) to determine when the project *would have finished* "but for" those events²². This method is entirely dependent on the quality and logic of the final as-built schedule.

A critical commercial issue in EOT claims is the treatment of concurrent delay. True concurrent delay occurs when an Employer Risk Event and a Contractor Risk Event happen at the exact same time, and both independently affect the critical path, causing a delay to completion²⁴. Under the SCL Protocol and general FIDIC jurisprudence, the standard treatment is that the Contractor is granted the Extension of Time (shielding it from the application of Liquidated Damages) but is not entitled to recover financial prolongation costs for that specific concurrent period, as the contractor would have been delayed and incurring overhead costs anyway due to its own culpable delay²⁴.

A formal EOT submission requires a highly structured narrative architecture. A comprehensive submission contains: (1) Executive Summary outlining the event and total days claimed; (2)

Statement of Facts providing a chronological detailing of the event; (3) Contractual Basis of Entitlement linking the facts to Sub-Clause 8.4 and demonstrating Sub-Clause 20.1 compliance; (4) Delay Analysis Methodology explaining the TIA or Windows approach; (5) Chronological Schedule Analysis detailing the critical path movement pre- and post-fragnet insertion; (6) Quantum Analysis detailing requested prolongation costs such as extended site overheads and insurance premiums; and (7) Appendices housing the contemporaneous records. These records must include daily site reports, localized weather data, Joint Measurement Records, formal correspondence, drawing registers, and minutes of meetings. The CLAIMS agent must maintain a dynamic Delay Event Log tracking the notification dates, clause references, and schedule impact of every disruption to prevent waiver of rights.

4. Running Account Billing

Running account billing, or progressive payment, forms the financial circulatory system of an EPC project, ensuring the contractor receives cash flow commensurate with the value of work executed rather than waiting for final project completion. Under a FIDIC Yellow Book solar EPC contract, the payment structure usually combines milestone payments (for major high-value procurement items) and progress-based payments (for civil and installation works). For instance, solar PV modules, tracking systems, and the 220kV autotransformers are typically paid upon successful factory acceptance testing (FAT), delivery to the site, and final installation¹⁶. Conversely, ground grading, pile driving for the MMS, and cable trenching are measured progressively based on the percentage of completion against the Contract Price breakdown.

The commercial cycle begins with the INVOICE agent's submission of the Payment Application (RA Bill) under Sub-Clause 14.3. This application is submitted at the end of each month and includes the estimated contract value of works executed, any approved variations, adjustments for changes in legislation or cost indices under Sub-Clause 13.8⁴, and mandatory deductions for retention money and advance payment recovery². The measurement and certification process requires the INVOICE agent to submit detailed measurement sheets, backed by approved Quality Assurance (QA) inspection requests, to the Engineer for verification. Within 28 days of receiving the Statement and all supporting documents, the Engineer is strictly obligated under Sub-Clause 14.6 to issue an Interim Payment Certificate (IPC) to the Employer, stating the amount the Engineer fairly determines is due⁶. Subsequently, under Sub-Clause 14.7, the Employer must pay the certified amount to the Contractor within 56 days of the Engineer receiving the Contractor's original Statement².

If the Employer fails to make payment within this 56-day timeframe, the Contractor has two primary, powerful remedies. First, under Sub-Clause 14.8, the Contractor is entitled to receive financing charges (interest) compounded monthly on the unpaid balance, calculated at the rate specified in the Appendix to Tender⁸. Second, under Sub-Clause 16.1, the Contractor may, after giving 21 days' formal notice to the Employer, suspend or reduce the rate of work until payment is received. If the suspension causes delay or incurs extra cost, the Contractor is entitled to an EOT and cost plus reasonable profit⁸.

Advance payment amortization is handled pursuant to Sub-Clause 14.2. A mobilization advance

(typically 10–15% of the Contract Price) is paid against an Advance Payment Guarantee. Recovery commences in the IPCs once the certified value of works reaches a predefined threshold (e.g., 10% of the Accepted Contract Amount) and is deducted at a set amortization rate (e.g., 25% of the gross amount of subsequent IPCs) until the advance is fully recovered⁸. Concurrently, the Performance Security, required under Sub-Clause 4.2, must be tracked rigorously. It must remain valid and enforceable until the Contractor completely executes the works and remedies any defects. If the guarantee features a fixed expiry date and the Performance Certificate has not been issued 28 days prior to that date, the Contractor must proactively extend the validity of the security to prevent an immediate, punitive encashment by the Employer⁵.

5. Subcontract Administration

Effective administration of utility-scale solar subcontracts relies heavily on the back-to-back principle, wherein the obligations, risks, and liabilities assumed by the main EPC contractor under the FIDIC Yellow Book are flowed down proportionally to the subcontractors. This ensures that the main contractor is not caught in a commercial or liability gap. Key provisions flowed down include equivalent design warranties, synchronized defects liability periods, identical specifications for materials, and mirrored dispute resolution mechanisms.

The CHECKER role is critical in reviewing and verifying subcontractor payment claims to prevent over-certification. Subcontractors for civil works (e.g., site leveling, foundation piling for the GSS) operate on item-rate contracts based on a localized Bill of Quantities. At the end of the billing cycle, the CHECKER and the subcontractor conduct a joint measurement on site. They calculate physical volumes and areas and record these on signed Joint Measurement Records (JMR). The CHECKER rigorously verifies that the measured work has passed quality control inspections before certifying the quantities for payment. To protect the main contractor's cash flow, subcontract agreements frequently incorporate "pay-when-paid" clauses, stipulating that the subcontractor is paid only after the main contractor receives corresponding funds from the Employer, subject to the legality of such clauses in the local jurisdiction.

Liquidated Damages (LDs) in the subcontract are designed to offset the main contractor's exposure to the Employer. If a delay by the electrical subcontractor in energizing the 33/220kV GSS triggers main contract LDs from the IPP developer, the main contractor passes these damages down to the culpable subcontractor. The subcontract LD rate is typically calibrated to cover both the Employer's LDs and the main contractor's extended site overheads.

Subcontractor variations must be managed with extreme caution. A subcontractor can only claim a variation if the change stems directly from an instruction issued by the main contractor. If the variation originates from the ultimate Employer (e.g., NEGU altering the grid connection criteria at the 220kV Ishtihan substation¹⁶), the main contractor will process the subcontractor's variation claim back-to-back with the Employer. The subcontractor is strictly not granted financial approval for the variation until the Employer has formally approved the corresponding main contract VO.

Upon substantial completion of their specific scope, the subcontractor enters the Defects Liability Period (DLP), mirroring the main contract (typically 12 to 24 months). During this phase,

retention money (usually 5-10% of the subcontract value) is held to ensure defects are rectified. Final accounts are settled only after the issuance of the overall Performance Certificate, requiring the subcontractor to sign a comprehensive no-claims discharge and complete all outstanding snagging items.

6. Commercial Coordination

The LEDGE agent (Commercial Coordinator) orchestrates the administrative and logistical frameworks that keep the commercial team synchronized and compliant. Central to this role is the maintenance of the Commercial Correspondence Register. This log tracks every incoming and outgoing letter, categorizing them by subject, contractual impact, mandated response deadlines, and associated FIDIC clauses. This register ensures that 28-day notice periods for claims under Sub-Clause 20.1 or 21-day notice periods for suspension under Sub-Clause 16.1 are never breached, thereby preserving the contractor's legal entitlements¹¹.

Insurance administration represents a critical commercial risk item requiring continuous oversight. Under Sub-Clause 18.2, the project must carry Contractor's All Risk (CAR) insurance covering physical damage to the permanent works, plant, and materials from commencement until taking over⁴. Third-Party Liability (TPL) insurance protects against damage to external property or injury to non-project personnel. Workmen's Compensation insurance is mandatory for all site labor. For an EPC project in landlocked Central Asia, Marine Cargo insurance is absolutely vital, as highly valuable solar PV modules and tracking systems transit internationally via rail and truck. The LEDGE agent ensures all policies remain valid through the DLP and that any claims (e.g., transit damage to tier-1 modules) are filed strictly within the policy's notification timeframes¹⁵.

Tracking bank guarantees is equally crucial. The Advance Payment Guarantee must match the unamortized balance of the advance, while the Performance Guarantee (typically 10% of the Contract Price) must remain valid until 28 days after the issuance of the Performance Certificate⁵. Retention Money Guarantees allow the contractor to release cash retentions early by providing equivalent bank security. The LEDGE agent maintains a digital register tracking all expiration dates, triggering internal renewal requests 45 days prior to expiry to prevent the Employer from making a punitive, on-demand call on the bond⁵.

Under FIDIC Sub-Clause 1.3, any formal Notice must be in writing, delivered by hand against receipt, sent by mail or courier, or transmitted via specifically agreed electronic systems, to the exact address specified in the Appendix to Tender⁴. Crucially, international courts and arbitral tribunals strictly enforce the principle that a document must clearly identify itself as a "Notice" and explicitly reference the specific contractual clause to be legally binding as a condition precedent⁴.

7. Dispute Management

The FIDIC Yellow Book 1999 cascades dispute resolution through a tiered, sequential process designed to prevent escalation and encourage early settlement. Clause 20 begins with the Dispute Adjudication Board (DAB). If the Engineer's Sub-Clause 3.5 determination is rejected by either party, the dispute may be referred in writing to the DAB. The DAB has 84 days to issue a written decision, which is binding on both parties and must be implemented immediately, even

if a party remains deeply dissatisfied with the outcome⁹.

If a party disagrees with the DAB's decision, it must issue a Notice of Dissatisfaction within 28 days of receiving the decision. This formal notice prevents the DAB decision from becoming "final and binding" and preserves the party's right to pursue arbitration⁹. Following the issuance of a Notice of Dissatisfaction, the parties must attempt Amicable Settlement for a mandatory period of 56 days before arbitration can commence. If amicable settlement fails, the dispute proceeds to final, binding resolution via international arbitration, typically governed by the Rules of Arbitration of the International Chamber of Commerce (ICC).

Disputes are best avoided through proactive, impeccable contract administration. Utilizing early warning mechanisms, maintaining a pristine contemporaneous record of site events, and raising technical queries instantly ensures that ambiguities in the Employer's Requirements are resolved before they compound into entrenched financial disputes.

8. Real Document Templates — Full Detail

The following templates represent the rigorous, standard formats required for the HELIOS Project, incorporating realistic data indicative of a utility-scale solar EPC in Uzbekistan.

8.1 Variation Register

VO No.	Date Issued	Ref / Notice No.	Description of Variation	Clause Ref	Proposed Value (USD)	Certified Value (USD)	EOT Claimed	EOT Granted	Status
VO-001	12-Feb-26	NTC-014	Rerouting 33kV cable to avoid unmapped utility line	13.1	\$45,200.00	\$40,000.00	0	0	Approved
VO-002	05-Mar-26	NTC-022	Upgrade GSS Auto-Transformer	13.1	\$320,500.00	\$320,500.00	14	14	Approved

			er from 100M VA to 125M VA						
VO-0 03	18-A pr-26	NTC- 039	Addit ional grou nd gradi ng at Block C	13.3	\$88,0 00.0 0	\$75,0 00.0 0	4	0	Appr oved
VO-0 04	22-M ay-26	NTC- 045	Chan ge in mod ule moun ting pile dept h due to Seis mic code	13.3	\$112, 300. 00	Pendi ng	12	Pendi ng	Unde r Revie w
VO-0 05	10-J un-2 6	NTC- 055	SCA DA syste m com muni catio n proto col upgr	13.1	\$25,0 00.0 0	\$25,0 00.0 0	0	0	Appr oved

			ade						
VO-006	15-Jul-26	NTC-061	BNEF Tier 1 Module Customs clearance alternative sourcing	13.7	\$15,000.00	\$0.00	8	0	Rejected
VO-007	02-Aug-26	NTC-072	Additional O&M Building HVAC capacity	13.1	\$18,400.00	\$17,000.00	0	0	Approved
VO-008	14-Sep-26	NTC-088	NEG U grid interconnection 220kV OHL realignment	13.1	\$210,000.00	Pending	21	Pending	Under Review
VO-009	20-Oct-26	NTC-095	Perimeter fencing	13.1	\$55,600.00	\$50,000.	0	0	Approved

			ng speci ficati on upgr ade to anti- climb		0	00			
VO-0 10	05-N ov-26	NTC- 101	CCT V came ra additi onal pole install ation s (15 units)	13.1	\$32,0 00.0 0	Pendi ng	3	Pendi ng	Unde r Revie w

8.2 Variation Order Form

VARIATION ORDER NO: HELIOS-VO-002 **DATE:** 05-Mar-2026 **PROJECT:** HELIOS 100 MW Solar PV + 33/220kV GSS **CONTRACTOR:** EPC Global Solutions **TO:** Project Manager, EPC Global Solutions

1. Description of Variation: In accordance with Sub-Clause 13.1 of the FIDIC Yellow Book 1999, the Contractor is hereby instructed to execute the following variation: Upgrade the primary 33/220kV step-up autotransformer capacity from 100MVA to 125MVA as per revised NEGU grid connection specifications attached.

2. Adjustment to Contract Price: Original Contract Price: \$85,000,000.00 Net Change by previous VOs: \$40,000.00 Amount of this VO: \$320,500.00 Revised Contract Price: \$85,360,500.00

3. Adjustment to Time for Completion: Extension of Time granted: 14 Days. Revised Completion Date: 12-Nov-2026.

Signed by the Engineer: _____ Date: 05-Mar-2026 Accepted by Contractor: _____ Date: 06-Mar-2026

8.3 EOT Submission Structure

Section 1: Executive Summary Overview of the delay event (NEGU route access denial), total EOT requested (21 days), and summary of cost impact (\$105,000).

Section 2: Statement of Facts Chronological detailing of the event. On 01-Sep-2026, local authorities, acting on behalf of the Employer, denied access to the approved routing for the 220kV Transmission Line. Access was finally granted on 22-Sep-2026.

Section 3: Contractual Entitlement Reference to Sub-Clause 8.4 (e) (Delay attributable to the Employer) and Sub-Clause 2.1 (Right of Access to the Site). Demonstration of compliance with Sub-Clause 20.1 (Initial notice issued on 10-Sep-2026).

Section 4: Delay Analysis Methodology Explanation of the Time Impact Analysis (TIA) deployed. Identification of the baseline schedule utilized (Update Rev 4, Data Date 01-Aug-2026).

Section 5: Schedule Analysis Details of Fragnet FR-004 inserted into the schedule. Demonstration of the critical path shift from the MMS installation to the OHL pylon foundation works.

Section 6: Quantum Assessment Prolongation costs requested, calculated strictly based on time-related site overheads, extended equipment hire, and insurance premium extensions during the 21-day delay period.

Section 7: Appendices

- App A: Sub-Clause 20.1 Notices and Correspondence
- App B: Schedule Update Rev 4 (Native XER & PDF)
- App C: Impacted Schedule (Native XER & PDF)
- App D: Site Diaries and Labor Logs

8.4 Delay Event Log

Event ID	Description	Notified Date	Clause	Actual Start	Actual Finish	Fragnet ID	Critical Impact
DLY-01	Unforeseeable hard rock in Block A	15-Jan-26	4.12	10-Jan-26	28-Jan-26	FR-01	5 Days
DLY-02	Late approval of Inverter Design	02-Mar-26	8.4(e)	15-Feb-26	10-Mar-26	FR-02	0 Days (Float)
DLY-03	Customs hold - BNEF Tier 1 cert	12-Jul-26	8.4(d)	01-Jul-26	20-Jul-26	FR-03	19 Days

	issue						
DLY-04	NEGU OHL Route Access Denial	10-Sep- 26	8.4(e)	01-Sep- 26	22-Sep- 26	FR-04	21 Days
DLY-05	Excepti onally adverse rainfall	05-Nov -26	8.4(c)	01-Nov -26	06-Nov -26	FR-05	4 Days

8.5 Payment Application (RA Bill)

RA Bill No: 08 **Period Ending:** 31-Aug-2026 **Contractor:** EPC Global Solutions

BoQ Ref	Descri ption	Unit	Total Qty	Rate (USD)	Prev Qty	Curren t Qty	Cum. Amount (USD)
2.1.1	Site Grading & Leveling	Ha	150	2,500.0 0	100	20	\$300,0 00.00
3.2.1	Supply of PV Module s (Tier 1)	MW	100	180,00 0.0	40	30	\$12,600 ,000.0 0
4.1.2	MMS Pile Driving	Nos	45000	45.00	20000	12000	\$1,440, 000.00
5.1.1	220kV Auto-tr ansfor mer	Ls	1	850,00 0.0	0.8	0.2	\$850,0 00.00

Gross Value of Works Executed							\$15,190,000.00
--------------------------------------	--	--	--	--	--	--	------------------------

8.6 Payment Certificate

Interim Payment Certificate No: 08 Date: 14-Sep-2026

Description	Amount (USD)
1. Gross Value of Works Executed	\$15,190,000.00
2. Approved Variations (Cumulative)	\$360,500.00
3. Total Gross Value (1+2)	\$15,550,500.00
4. Less: Advance Payment Amortization (15%)	(\$2,332,575.00)
5. Less: Retention Money (5%)	(\$777,525.00)
6. Net Amount Certified this Period	\$12,440,400.00
7. Less: Previous Payments Certified	(\$8,200,000.00)
8. Net Amount Payable	\$4,240,400.00

8.7 FIDIC Notice Form

NOTICE NO: HELIOS-NTC-088 **TO:** The Engineer, [Firm Name] **DATE:** 10-Sep-2026

REFERENCE CLAUSE: Sub-Clause 20.1 (Contractor's Claims) & Sub-Clause 8.4(e)

SUBJECT: Notice of Claim for Extension of Time and Additional Cost due to Denial of Access to 220kV OHL Route

NARRATIVE: In accordance with Sub-Clause 20.1 of the FIDIC Yellow Book 1999, the Contractor hereby gives formal notice that it considers itself entitled to an Extension of Time and additional payment under Sub-Clause 8.4(e). On 01-Sep-2026, the Contractor's

mobilization team was denied access to the approved routing for the 220kV Transmission Line by local authorities, contrary to the Employer's obligation to provide right of access under Sub-Clause 2.1. This delay is currently impacting critical path activities. A fully detailed claim will be submitted within 42 days of the event's commencement as required by the Contract.

8.8 Subcontractor Measurement Sheet (JMR)

JMR NO: CIV-JMR-014 **SUBCONTRACTOR:** Local Earthworks LLC **WORK ITEM:** Inverter Station Foundation Concrete (Item 4.3.2) **LOCATION:** Block B, Pad 1-4

Ref	Location	Length (m)	Width (m)	Depth (m)	No. of Units	Volume (m ³)	Remarks
1	Pad B1	6.0	4.0	0.5	1	12.0	Pour complete 10-Aug
2	Pad B2	6.0	4.0	0.5	1	12.0	Pour complete 11-Aug
3	Pad B3	6.0	4.0	0.5	1	12.0	Pour complete 12-Aug
4	Pad B4	6.0	4.0	0.5	1	12.0	Pour complete 13-Aug
Total Volume for Certification:						48.0 m³	Approved by CHECKER

8.9 Subcontractor Payment Certificate

SUBCONTRACTOR: Local Earthworks LLC **IPC NO:** Sub-04 **DATE:** 05-Sep-2026

Description	Amount (USD)
Gross Value of Works	\$120,000.00
Less: Retention (5%)	(\$6,000.00)
Less: Advance Recovery (10%)	(\$12,000.00)
Less: Back-charges (Fuel/Equipment)	(\$1,500.00)
Net Amount Payable	\$100,500.00

8.10 Commercial Risk Register

Risk ID	Description	Probability	Financial Impact	Schedule Impact	Mitigation Strategy	Owner
R-01	Delayed Employer Payment	High	Cash flow negative	Potential suspension	Issue 14.8 interest notice, prep 16.1	COUNSELL
R-02	Customs rejects PV Modules	Medium	High storage fees	Critical delay	Ensure BNEF Tier 1 certs are pre-cleared	SCM
R-03	Subcontractor Insolvency	Low	Cost to complete	Moderate delay	Monitor sub labor payments , hold retention	CHECKER
R-04	Concurrent rain	High	Prolongation lost	0 (EOT only)	Optimize schedule	CLAIMS

	delay				logic to separate delays	
--	-------	--	--	--	--------------------------------	--

9. Contracts Interfaces

The Contracts Manager does not operate in a commercial vacuum. Successful commercial execution relies on precise, continuous interfacing with multiple project departments:

- **With Planning:** The CLAIMS agent must continuously feed delay event data to the Planning team. Planning provides the native XER schedule files required for the Time Impact Analysis. Contracts relies implicitly on Planning to ensure the schedule baseline logic is sound so that fragnet insertions yield mathematically defensible EOT entitlements that withstand Engineer scrutiny.
- **With Construction:** The construction team is the primary source of contemporaneous records. Contracts interfaces with site managers to ensure that daily diaries explicitly note weather impacts, idle equipment, and instructions given verbally by the Engineer. Without construction documenting the physical reality, the CLAIMS agent has no evidentiary foundation to satisfy Sub-Clause 20.1.
- **With Supply Chain Management (SCM):** Contracts manages vendor back-charges if equipment arrives defective or late. Furthermore, any PO variations due to market volatility or customs regulation changes (e.g., Uzbekistan's new BNEF Tier 1 requirements restricting imports to specific manufacturers¹⁴) must be legally codified by Contracts to protect the project margin.
- **With the Client:** The COUNSEL serves as the primary formal conduit to the Employer and the Engineer. This involves submitting Sub-Clause 1.3 Notices, negotiating VO values in Sub-Clause 3.5 meetings, and persistently following up on overdue IPCs to ensure the project remains capitalized.
- **With Finance:** Delayed payments severely disrupt EPC cash flows. Contracts informs Finance of the expected dates for IPC realization and legally calculates Sub-Clause 14.8 financing charges on late payments. The LEDGE agent also instructs Finance when to renew Bank Guarantees to prevent lapses in validity that could trigger encashment⁵.

10. Common Contracts Problems

1. Client issues an instruction that appears to be out of scope If the Engineer issues a drawing or directive that the Contractor considers out of scope, the COUNSEL must not proceed blindly. The Contractor must immediately notify the Engineer under Sub-Clause 13.1 that the instruction constitutes a Variation, or issue a Sub-Clause 20.1 notice if it impacts time/cost. If the Contractor proceeds without issuing notice, the Engineer will subsequently argue the work was inherently included in the original lump-sum Contract Price.

2. Contractor misses the 28-day notice for a delay event As confirmed by the Privy Council in *UBC v WASA*, failing the 28-day notice under Sub-Clause 20.1 is generally fatal to the claim¹⁰.

If the deadline is missed, the Contractor's options are severely limited. The COUNSEL may argue that earlier informal correspondence (like a site meeting minute or progress report) constituted constructive notice, or argue that the Employer waived the requirement by actively assessing the claim anyway⁹. However, these are high-risk fallback legal arguments; the primary defense is strict administrative compliance.

3. Subcontractor abandons site mid-project If a civil subcontractor abandons the site, the CHECKER immediately freezes all pending payments and retention money. The COUNSEL formally terminates the subcontract for default. The unamortized advance payment bond and performance bond are immediately called. The site works are measured "as-is" via a unilateral JMR, and the SCM team procures a replacement contractor. All additional costs to complete the original scope are levied as a back-charge against the defaulting subcontractor's final account.

4. Client withholds payment without valid reason If an IPC is issued but not paid within the 56-day limit under Sub-Clause 14.7, the COUNSEL issues a formal demand for payment and notifies the Employer of the entitlement to financing charges under Sub-Clause 14.8⁸. If payment remains outstanding, the COUNSEL issues a 21-day notice under Sub-Clause 16.1 to suspend the works⁸. If the suspension continues for 84 days, it matures into grounds for termination by the Contractor under Sub-Clause 16.2.

5. Variation is orally instructed on site but not confirmed in writing Under Sub-Clause 3.3, if the Engineer gives an oral instruction, the Contractor must confirm it in writing within two working days. If the Engineer does not reject this written confirmation within a further two working days, the confirmation is legally deemed to be a formal written instruction from the Engineer. If the Contractor fails to issue this confirmation, recovering costs for the executed work becomes contractually precarious.

6. Two delay events overlap (concurrent delay) If an Employer Risk Event (e.g., late design approval of the GSS control building) and a Contractor Risk Event (e.g., inability to mobilize piling rigs) run concurrently and independently impact the critical path, the SCL Protocol dictates that the Contractor is entitled to an Extension of Time for the concurrent period²⁴. This relieves the Contractor from Liquidated Damages. However, the Contractor is typically barred from recovering prolongation costs for that specific overlapping period, as it cannot demonstrate that the Employer's delay was the sole cause of the financial loss²⁴. The CLAIMS agent must surgically partition the delay analysis using Windows Analysis to separate concurrent periods from sequential, compensable Employer delays.

Works cited

1. FIDIC Yellow Book 2017: Clause 8 Overview | PDF - Scribd, <https://www.scribd.com/document/756940047/8-FIDIC-Yellow-Book-2017>
2. FIDIC Yellow Book 1999: Clause 4.9 QA | PDF | License - Scribd, <https://www.scribd.com/document/912219349/FIDIC-Yellow-Book-1999-Design-and-Build>
3. FIDIC 4th Edition: A Practical Guide | PDF | Employment - Scribd, <https://www.scribd.com/document/741580302/A-practical-legal-guide>

4. روعة Fidic Red Book 1999 Clauses Risks - CORBETT&CO,
<https://pdfcoffee.com/-fidic-red-book-1999-clauses-risks-corbettampco-pdf-free.html>
5. Performance Security in Construction Contracts | PDF - Scribd,
<https://www.scribd.com/document/396507848/Performance-Security-Notes>
6. FIDIC Legal Disputes Overview 2023 | PDF - Scribd,
<https://www.scribd.com/document/665580691/Table-of-FIDIC-Cases-03-July-2023-update>
7. The Guide To Construction Arbitrator 3rd Edition | PDF - Scribd,
<https://www.scribd.com/document/442262715/The-Guide-to-Construction-Arbitrator-3rd-Edition>
8. FIDIC Yellow Book 2017 Clause 14 Overview | PDF | Taxes - Scribd,
<https://www.scribd.com/document/784176126/Clause-14-2017>
9. Variations Claims Under FIDIC: No Notice, No Payment? Lessons,
<https://internationalconstructionknowledgehub.com/variations-claims-under-fidic-no-notice-no-payment-lessons-from-ubc-v-wasa-2026-ukpc-2/>
10. FIDIC time bar bites – Privy Council holds clause 20.1 is a condition,
<https://www.charlesrussellspeechlys.com/en/insights/expert-insights/construction-engineering-and-projects/2026/fidic-time-bar-bites--privy-council-holds-clause-20.1-is-a-condition-precedent>
11. J.V. of American and EU Dredging Companies v. Red Sea Public,
<https://www.scribd.com/document/595624545/J-V-of-American-and-EU-Dredging-Companies-v-Red-Sea-Public-Authority-RSPA-Final-Award-CRCICA-Case-No-281-2002-28-June-2004>
12. Clause 13: Variations and Adjustments | PDF - Scribd,
<https://www.scribd.com/presentation/484649581/CLAUSE-13>
13. Limitation of Liability in Contracts | PDF | Damages - Scribd,
<https://www.scribd.com/document/453803119/Sub-Clause-1-15-Limitation-of-Liability-pdf>
14. The Ultimate Guide to Customs Clearance in Uzbekistan (2024,
<https://azma.uz/tpost/g9pcsxte1-the-ultimate-guide-to-customs-clearance>
15. Solar Panel Requirements in Uzbekistan: Permits & Grid Connection,
<https://eai.uz/en/articles/solar-panel-installation-requirements-uzbekistan>
16. Uzbekistan: Masdar 897MW Solar PV Portfolio (Samarkand, Jizzakh,
<https://www.aiib.org/en/projects/details/2023/approved/Uzbekistan-Masdar-897MW-Solar-PV-Portfolio-Samarkand-Jizzakh-and-Sherabad-solar-PV-plants.html>
17. Undergo customs clearance - Uzbekistan Trade Info,
<https://uztradeinfo.uz/procedure/425/step/913>
18. FIDIC - 1999 Red Book - Outline of Claim - Scribd,
<https://www.scribd.com/document/659535628/FIDIC-1999-Red-Book-Outline-of-Claim>
19. Delay Analysis Methods: The Six SCL Options Explained - Aven-AI,
<https://aven-ai.com/blog/delay-analysis-methods-scl-protocol>
20. A Foundational Reference in Delay Analysis - SCCSI,
<https://sccsinternational.org/scl-protocol-1/>

21. Modified Time Impact Analysis Method - ASCE Library,
<https://ascelibrary.org/doi/10.1061/%28ASCE%29CO.1943-7862.0000402>
22. Time Impact vs As-Planned vs As-Built Explained | CALIM,
<https://www.calim.ai/insights/delay-analysis-methods-explained-as-planned-vs-as-built-windows-and-impacted-programmes/>
23. Society of Construction Law - Williams Planning,
<https://willplan.co.nz/society-of-construction-law/>
24. SCL Delay and Disruption Protocol 2nd Ed. | PDF | Damages - Scribd,
<https://www.scribd.com/presentation/814353652/Scl-Delay-and-Disruption-Protocol-r3>
25. Specific Procurement Notice, <https://www.uzbekistonmet.uz/en/lists/view/7061>
26. FIDIC Contract Training Overview | PDF - Scribd,
<https://www.scribd.com/document/821444095/The-FIDIC-Form-of-Contract-April-24-2024-Final>
27. Presentation On Clause 16, Suspension & Termination by Contractor,
<https://www.scribd.com/presentation/779950942/Presentation-on-Clause-16-Suspension-Termination-by-Contractor>
28. Presentation On Clause 18, Insurances | PDF - Scribd,
<https://www.scribd.com/presentation/779948098/Presentation-on-Clause-18-Insurances>
29. J Murphy & Sons Ltd v Beckton Energy Ltd - CaseMine,
<https://www.casemine.com/judgement/uk/5a8ff76160d03e7f57eabecf>